Sharzil Hasnine

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Portfolio: shas-9.github.io

Education

Bachelor of Computer Science (Advanced) Feb 2023 – Nov 2026
Adelaide University

Experience

Industry Project - Block Model Compression Algorithm - Maptek Aug 2025 - Oct 2025
Tech: *C++, Bash, Jira*

- Architected a stream-oriented lossless compression system for 3D data, optimising for speed and compression ratio.
- Implemented an octree-based spatial compression algorithm, achieving 98.8% reduction in data size while improving processing speed by 2.6x over the uncompressed baseline, forming the foundation for subsequent team optimisations reaching 99.17% compression and 1520x processing speed.
- Designed a testing framework that verifies correctness of compressed outputs against uncompressed input data, ensuring algorithmic correctness independent of compression strategy and enabling automated, extensible testing.
- Built a cross-platform CI pipeline with automated testing and reproducible Linux and Windows builds, integrating cross-compilation and artifact generation into the development workflow, improving team velocity by 1.5x.

Technical Volunteer - Autonomous navigation - Adelaide Rover Team Oct 2024 - Feb 2025
Tech: *Python, C++, Ros2, Nav2, Gazebo, Groot*

- Designed a behavior tree for the rover's autonomous navigation pipeline using Nav2, enabling modular decision-making for tasks such as path planning, recovery behaviours, and goal execution.
- Simulated navigation behaviour using Gazebo, supporting the Adelaide Rover Team's preparation for the Australian Rover Challenge's autonomous navigation task.

Projects

CatQuest – 2D Maze Exploration Game Aug 2023 – Oct 2023
Tech: C++, SFML
Link: *github.com/Shas-9/Labyrinth-game*

- Developed gameplay systems for a real-time 2D maze exploration game in C++ using SFML, working with the main game loop, player state and interactive entities.
- Implemented enemy behaviour, player mechanics, collision detection and health systems while integrating them with an existing procedurally generated maze.
- Designed modular game-state and UI architecture to support menu navigation and transitions between different gameplay states without disrupting the real-time game loop.
- Debugged interactions between gameplay systems and refined implementation details to maintain predictable behaviour during real-time execution.

Monkey Run – Unihack 2024 Mar 2024
Tech: Godot, GDScript, Python, MediaPipe
Link: *github.com/rayokamoto/unihack*

- Developed an endless-runner game in Godot using GDScript, including gameplay logic and randomised obstacle generation.
- Implemented a Python and MediaPipe input system that translated real-time webcam-tracked player movement into game controls.
- Integrated the external input system with gameplay mechanics, debugging interactions between multiple components to deliver a responsive interactive experience.

Technical Skills

Languages: C++, Java, Typescript, Javascript, Python, C, GDScript, Matlab, Bash
Libraries and Frameworks: Node.js, Springboot, MongoDB, MySql, React, JUnit, Godot, Pandas, NumPy, Matplotlib, Sklearn, Tensorflow
Tools and Software: Git, Linux, Docker, Jira